

Lachlan Black

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Education

Duke University, Durham, NC

Expected May 2028

B.S. in Computer Science, Innovation & Entrepreneurship Certificate

Relevant Coursework: Computer Architecture, Data Structures and Algorithms, Data Science

Work Experience

Software Engineer Intern | GRACE Lab | Durham, NC

May 2026 – Jul 2026

- Engineered high-throughput telemetry processing pipelines with Polars and Xarray to reduce data latency by 60%.
- Trained and optimized deep learning models (LSTMs, PINNs) in PyTorch for renewable energy forecasting.
- Containerized multi-stage training environments via Docker to minimize production environment setup time.
- Collaborated within a 4-person team to deliver key model findings to be presented at the NASA POWER summit.

Technology Scout | Christensen Family Center for Innovation | Durham, NC

Jan 2026 – May 2026

- Evaluated the technical architecture of faculty and PhD venture concepts to assess viability for commercial deployment.
- Analyzed emerging software frameworks across university research labs to identify high-potential technology transfers.

Projects

Distributed Fleet Telemetry & Streaming Pipeline

Jul 2026 – Present

- Developed an event-driven Python and Apache Kafka pipeline processing 15,000+ events/sec across partitioned topics.
- Implemented stateful sliding-window aggregations and anomaly detection to identify battery and descent faults in <50ms.
- Configured multi-consumer groups with manual offset commits and DLQs to ensure fault tolerance and zero data loss.
- Engineered a FastAPI WebSocket gateway and Redis cache to stream telemetry to a live troubleshooting dashboard.

Autonomous UAV & Flight Controller

Jan 2026 – Jun 2026

- Built a 1 kHz C++ flight loop on a Teensy 4.0 executing real-time PID control algorithms to stabilize a carbon-fiber UAV.
- Developed a two-state Kalman filter to fuse barometer and IMU data for altitude and vertical velocity estimation.
- Programmed BMI323 registers to apply a 10 Hz low-pass filter to isolate true motion from motor vibrations.

Remote Sensing Data Pipeline

Oct 2025 – Nov 2025

- Implemented multi-threaded Dask workflows to run parallel matrix reductions on 50GB+ geospatial array datasets.
- Designed a partitioned PostgreSQL schema with spatial indexing to accelerate trend-modeling and analytical queries.
- Engineered an asynchronous FastAPI serialization layer to stream geospatial payloads and cut I/O latency by over 35%.

Leadership and Involvement

Undergraduate Teaching Assistant | Duke Computer Science | Durham, NC

Jan 2026 – May 2026

- Mentored 100+ students in core CS concepts, debugging workflows, and algorithmic complexity analysis.

Skills

Programming Languages: Python, C, C++, TypeScript/JavaScript, SQL

Libraries & Frameworks: PyTorch, FastAPI, React, Next.js, Polars, Xarray, Dask, pandas

Databases & Infrastructure: PostgreSQL, Apache Kafka, Redis, Docker, Git, Linux